AQA 3.1.12 Acids and bases

AQA A-level Chemistry · Revision questions · Sheet 7 of 8

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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21 marks · about 25 minutes

Key information

- $pH = -\log_{10}[H^+]$ and $[H^+] = 10^{-pH}$
- $K_w = [H^+][OH^-]$ (1.00×10^{-14} mol^2 dm^{-6} at 25 °C); for a base $[H^+] = K_w \div [OH^-]$
- Weak acid on its own: $[H^+] = \sqrt{(K_a \times [HA])}$; $K_a = 10^{-pKa}$
- Buffer: $[H^+] = K_a \times [HA] \div [A^-]$ — use moles, the volume cancels
- Acid added to a buffer: $A^- + H^+ \rightarrow HA$, so mol HA goes up and mol A^- goes down by the same amount

Section A: Short-answer questions

01 Values of the ionic product of water (K_w) at different temperatures are given.

$K_w = 6.40 \times 10^{-15}$ mol^2 dm^{-6} at 18 °C $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 25 °C

Which statement is correct?

[1 mark]

A The concentration of hydroxide ions in water at 18 °C is 8.00×10^{-8} mol dm^{-3}

B The dissociation of water into ions is an exothermic process.

C The pH of water is the same at 25 °C and at 18 °C

D Water becomes less acidic as the temperature is raised.

02 Calculate the pH of a 2.00×10^{-3} mol dm^{-3} solution of potassium hydroxide at 25 °C. Give your answer to 2 decimal places. $K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6}

[2 marks]

03 Which is the correct expression for the ionic product of water, K_w?

[1 mark]

A $K_w = [H_2O] \div ([H^+][OH^-])$

B $K_w = [H^+][H_2O]$

C $K_w = [H^+][OH^-]$

D $K_w = [H^+][OH^-] \div [H_2O]$

04 Suggest two substances that could be mixed in solution to make a basic buffer.

[1 mark]

05 Which is the concentration of NaOH(aq), in mol dm^{-3}, that has pH = 14.30? ($K_w = 1.00 \times 10^{-14}$ mol^2 dm^{-6} at 25 °C)

[1 mark]

A −1.16

B 5.01×10^{-15}

C 2.00×10^{14}

D 2.00

06 Calculate the pH of a 0.200 mol dm^{-3} solution of methanoic acid. Give your answer to 2 decimal places.
$K_a = 1.78 \times 10^{-4}$ mol dm^{-3}

[2 marks]

Section B: Longer questions

07 This question is about propanoic acid, a weak acid.

07.1 Give the expression for the acid dissociation constant (K_a) for propanoic acid (CH_3CH_2COOH).

[1 mark]

07.2 Calculate the pH of a 0.100 mol dm^{-3} propanoic acid solution. Give your answer to 2 decimal places.
For propanoic acid, $pK_a = 4.87$

[4 marks]

08 This question is about buffer solutions made from ethanoic acid.

08.1 State how a buffer solution can be made from solutions of potassium hydroxide and ethanoic acid.

Give an equation for the reaction between potassium hydroxide and ethanoic acid.

State how this buffer solution resists changes in pH when a small amount of acid is added.

[3 marks]

08.2 A buffer solution is made by adding 2.00 g of sodium hydroxide to 500 cm^3 of 1.00 mol dm^{-3} ethanoic acid solution.

Calculate the pH of this buffer solution at 25 °C. Give your answer to 2 decimal places. For ethanoic acid, $K_a = 1.74 \times 10^{-5}$ mol dm^{-3} at 25 °C

[5 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	**A**	*Source: 7405/3 Jun 2024 Q18*	1
02	M1 $[H^+] = 1.00 \times 10^{-14} \div 2.00 \times 10^{-3} = 5.00 \times 10^{-12}$ M2 pH = 11.30	*Source: Practice question (written for this set)*	2
03	**C**	*Source: 7405/1 Jun 2017 Q03.1*	1
04	Ammonia **and** ammonium chloride (or another ammonium salt)	*Source: Practice question (written for this set)*	1
05	**D**	*Source: 7405/3 Nov 2020 Q32*	1
06	M1 $[H^+] = \sqrt{(1.78 \times 10^{-4} \times 0.200)} = 5.97 \times 10^{-3}$ M2 pH = 2.22	*Source: Practice question (written for this set)*	2
07.1	$K_a = [H^+][CH_3CH_2COO^-] \div [CH_3CH_2COOH]$	*Source: 7405/1 Jun 2023 Q10.2–10.3* Square brackets essential	1
07.2	M1 $K_a = 10^{-pKa} = 1.35 \times 10^{-5}$ M2 $K_a = [H^+]^2 \div [CH_3CH_2COOH]$ M3 $[H^+] = \sqrt{(1.35 \times 10^{-5} \times 0.1)} = 1.16 \times 10^{-3}$ mol dm^{-3} M4 pH = 2.94	Allow 2.93	4
08.1	M1 add excess ethanoic acid to KOH / add enough KOH so the acid is only partially neutralised M2 $KOH + CH_3COOH \rightarrow CH_3COOK + H_2O$ M3 CH_3COO^- (from salt) reacts with (added) acid / H^+	*Source: 7405/1 Jun 2022 Q04.4–04.5* M2 allow $CH_3COO^-\ K^+$ M3 ignore equilibrium shifts	3
08.2	M1 n(NaOH) = 2.00 ÷ 40.0 = 0.05 mol **and** $n(CH_3COOH)$ = 0.5 mol M2 $n(CH_3COOH)$ after = 0.50 – 0.05 = 0.45 mol M3 $n(CH_3COO^-)$ = n(NaOH) = 0.05 mol M4 $[H^+] = 1.74 \times 10^{-5} \times 0.45 \div 0.05 = 1.57 \times 10^{-4}$ M5 pH = 3.80	V cancels so is not needed M5 allow 3.81; Henderson–Hasselbalch method allowed	5